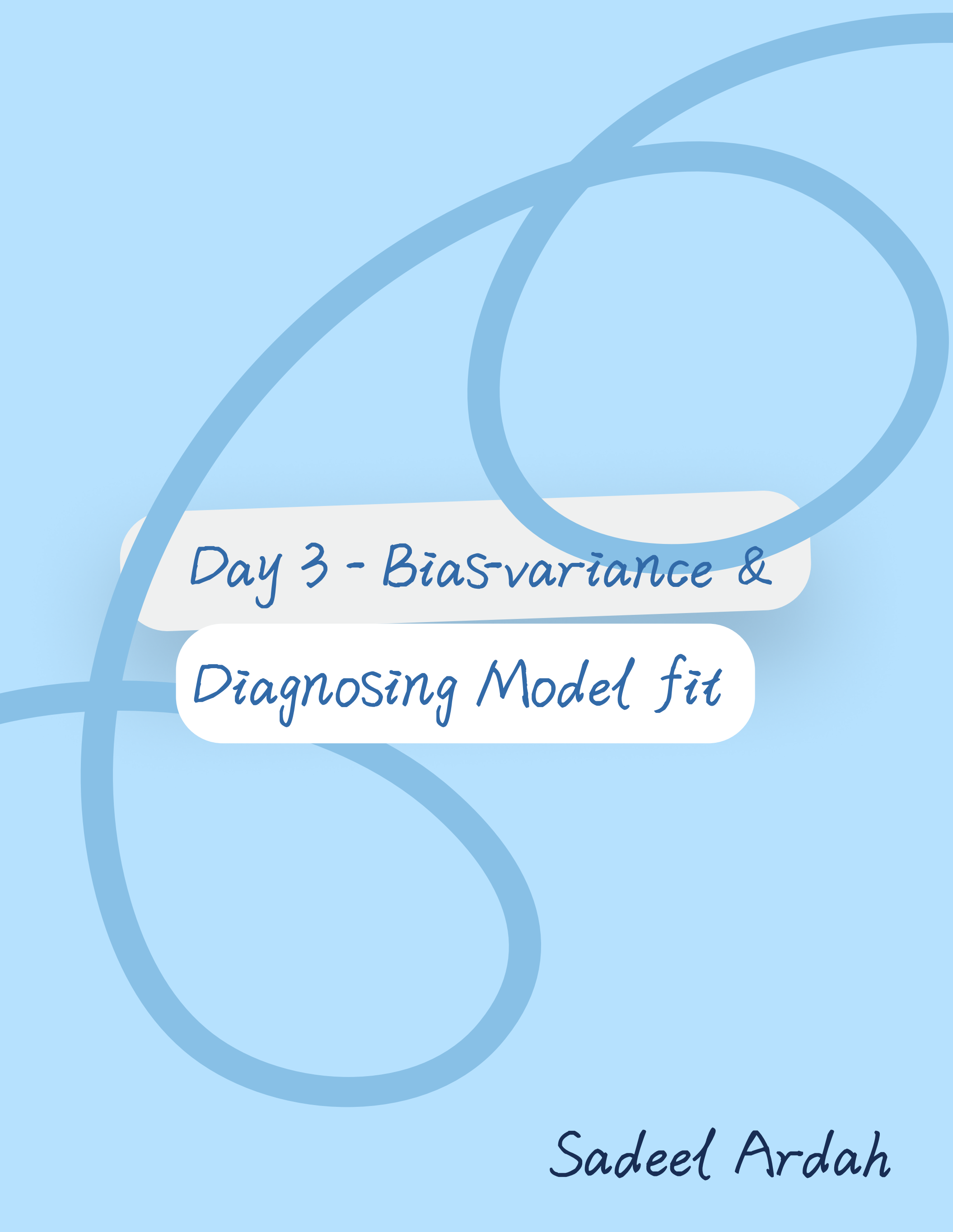




Day 3 - Bias-variance &

Diagnosing Model fit

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Bias-variance & Diagnosing Model fit

Learning objectives:-

- 3.1 — The two ways a model fails ✓ overfitting ✓ underfitting ✓
- 3.2 — The Bias-variance trade-off ✓
- 3.3 — Diagnosing with the train-vs-validation Gap ✓
- 3.4 — Regularization ✓

The Question of today: Why is my model performing badly, and how do I diagnosing the problem?

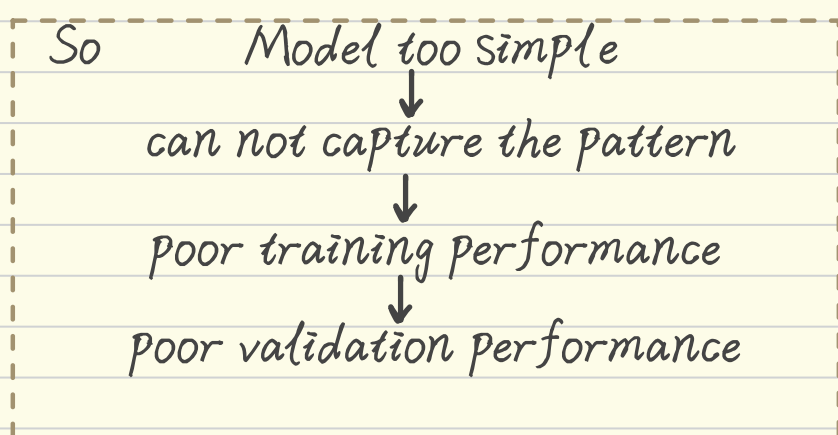
model failures are usually one of two type, the problem is that the treatment of the second so the first thing we need to do.

Diagnose the problem → choose the fix

Case one : Underfitting or High Bias

The Model is too simple so it is unable to learn the pattern found in the data, Imagine we have a complex problem, but the Model is trying to solve it in a very simple way

So the result :- Training score → poor } because the model originally didn't learn
validation score → poor } Training data well



Symptom:

Poor on both training AND validation data

Cause:

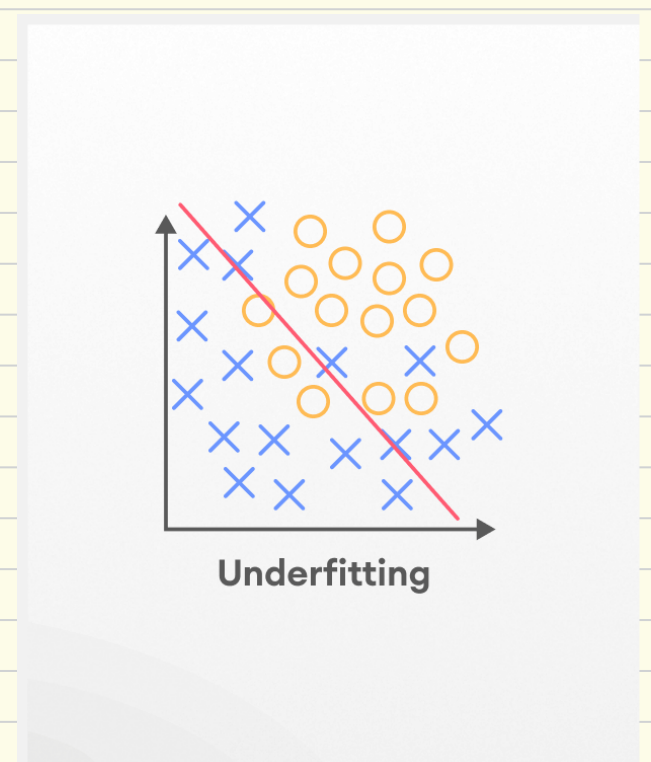
Model too simple to capture the pattern

Fix:

Add features

or

use a more powerful model

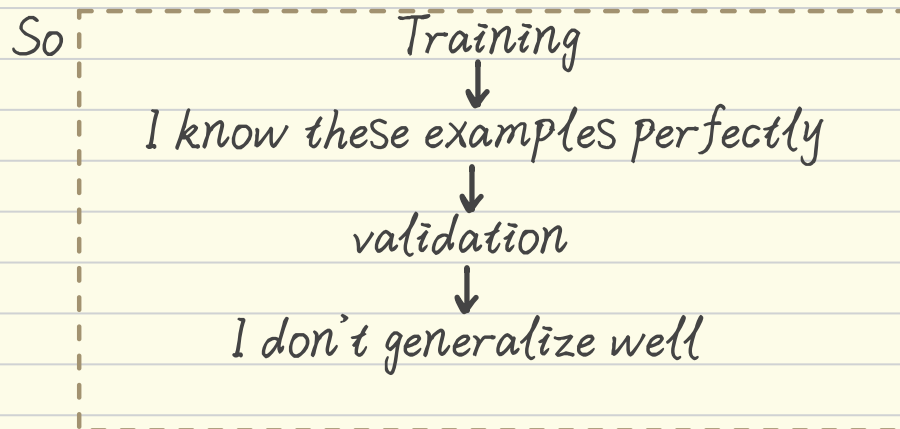


Case two : overfitting or High variance

The model become very complicated, Instead of learning the general pattern, he starts memorizing the details of training data so the result.

Training Score → Great
validation Score → Poor

*The model is excellent on the data it has been trained on, but it is bad for what it shows data it has not been trained on.



Symptom:

Great on training
Poor on validation

Cause:

Model too complex
Memorized the training data

Fix:

Simplify the model
Add data
Regularize

Note: - To judge the model should look at the training score and validation score

if the Training score low and the validation score low the model underfitting

if the Training score High and the validation score low the model overfitting

if the Training score and the validation score approximately equal the model Good fit

later ←

Bias & Variance

Bias → is the error caused by oversimplified assumptions

variation → is the error caused by the model's high sensitivity to the specified training data

First: Bias

Bias often related with the model too simple.

It means that the model has very simple assumptions and doesn't have enough ability to learn the real pattern.

For example:- Real Pattern = complex
Model = very simple

} means — Training Score Low
Validation Score low

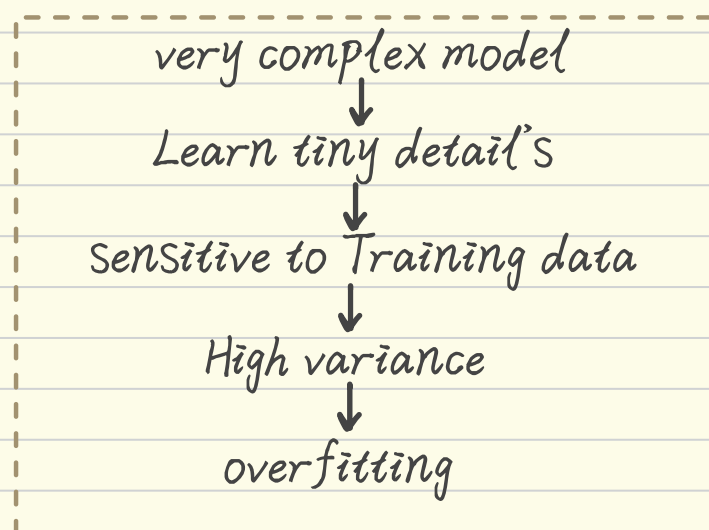
So the result: High bias → underfitting

High Bias
→ model too much simple
→ underfitting

Second: variance

variance often is related the with sensitivity of the model to training data.

It means that if the model becomes too complicated it can greatly influenced by the fine details found in Training Data So the situation (case) becomes:



means :
Trainig Score: very high
validation Score: much lower

High variance
→ model too much complex
→ overfitting

Trade off

As we make the model more complex the bias goes down but the variance goes up for example :

Model complexity Low $\longrightarrow$ High
Initial the model is very simple.
High bias
low variance

we complicate it a little more. -

Low bias $\nearrow$ that's good because he's able to learn the pattern better but if we keep going
High variance $\longleftarrow$
It's starts overfitting

Model too simple

Complexity = Low

Bias = High
Variance = Low

Result:
Underfitting ✗

The model is not able to learn the data natively

Balanced complexity

Complexity = Moderate

Bias = Reasonable
Variance = Reasonable

Result:
Good Fit ✓

The model can learn the real pattern
Generalize well

Model too complex

Complexity = High

Bias = Low
Variance = High

Result:
Overfitting ✗

The model Learned training data too much

Too Simple

Underfitting

High Bias

Sweet Spot

Good Fit

Balanced

Too Complex

Overfitting

High Variance

$\longrightarrow$ Model Complexity $\longrightarrow$

Train vs validation Gap

The first case

Training Score = low } Underfitting cause the model is weak even
Validation Score = low } on training data (model too simple)

The second case

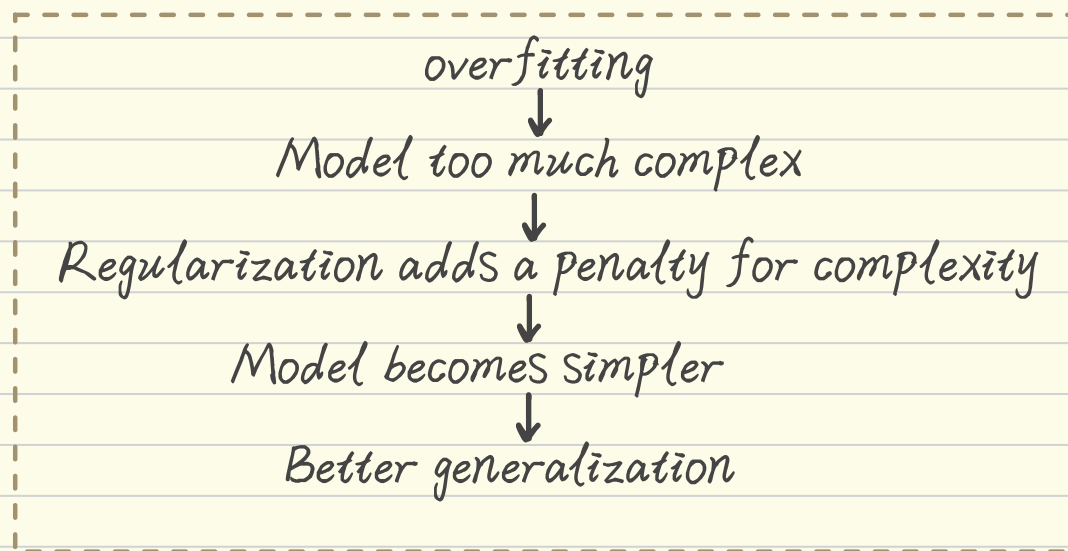
Training Score = High } overfitting cause the model is excellent on
Validation Score = much lower } training data, but dose not generalize well
(Large Gap)

The third case

Training Score = High } Good fit (small Gap) plus generalize well
Validation Score = High }

Training Score	Validation Score	Diagnosis
Low	Low	Underfitting — model too simple
High	Much lower	Overfitting — model memorized training data
High	High, small gap	Good fit — this is the target

Regularization



Instead of letting the model rely too heavily on certain features, Regularization reduces this dependence

There is two type of linear models :

- ↳ Ridge = L2
- ↳ Lasso = L1

1. Ridge L2

python Code

```
from sklearn.linear_model import Ridge  
ridge = Ridge(alpha = 1.0)
```

Ridge miniaturizes the weights towards zero

means:

```
graph TD; A[Large weights] --> B[Penalty]; B --> C[smaller weights];
```

2. Lasso

python Code

```
from sklearn.linear_model import Lasso  
lasso = Lasso(alpha = 1.0)
```

Lasso also reduce weights, but it has in additional advantage : It is possible to completly abandon some weights = 0

Practically possible to operate some kind of feature selection

↳ Because the feature weighed zero, as if the model ignored it

Alpha

It contro the power of the penalty

means: Large alpha
↓
Stronger Penalty
↓
Simpler model

For example:

-Ridge (alpha = 0.1) → Regulation is light

-Ridge (alpha = 10.1) → Regulation is stronger

But ; too little regularization → overfitting may remain
too much regularization → model may become too simple
underfitting

Regularization

- reduces model complexity
- helps fight overfitting

Ridge (L2)

- shrinks weights toward zero

Lasso (L1)

- can shrink some weights exactly to zero

alpha

- controls penalty strength

larger alpha

- simpler model

الحمد لله حمداً كثيراً طيباً مباركاً فيه